Zero-Knowledge Zakat: Privacy, Transparency, and Sharia Compliance on the Blockchain

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***Abstract*—Zakat is a mandatory annual obligation in Islam for wealth redistribution. Blockchain-based zakat platforms have emerged, but existing systems log complete donor-recipient transactions on public ledgers, contradicting the Islamic principle of *nafs* (dignity). This paper presents a protocol that combines off-chain ZK proofs, nullifier-based double-donation prevention, and selective disclosure. Donors select full anonymity or partial disclosure while receiving SBT receipts. On-chain verification provides institutional transparency, but individual transaction data stays private when the donor selects Private mode and the client-side proving pipeline has not been compromised. A Sepolia prototype generates UltraHONK proofs in 246.8 ms (Barretenberg v4.2.1, 16-core Linux) with end-to-end `donateZK()` consuming 721,345 gas. Cryptographic privacy and institutional accountability prove compatible, supporting digital zakat that preserves donor and recipient privacy. The protocol contributes a privacy layer for Islamic social finance that directly addresses nafs protection under maqasid al-shariah.**

***Index Terms*—Barretenberg, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Pedersen Hash, Privacy-Preserving Donations, Sharia Compliance, Soulbound Token, Zakat, Zero-Knowledge Proofs**

I. INTRODUCTION

BAZNAS [1] estimates Indonesia's annual zakat potential at Rp 327 trillion (approximately USD 20 billion), yet actual collection remains far lower. Blockchain-based zakat platforms have emerged to address operational gaps in collection and distribution. However, existing systems log complete donor-recipient transactions on public ledgers, exposing socioeconomic vulnerabilities that contradict Islamic privacy norms.

Maqasid al-shariah protects *nafs* (life and dignity) among its five essential objectives [5]. As Thoriquttyas and Rohmawati note, "Al-Shatibi does not clearly define human dignity (karāmah) as a separate maqṣad, unlike modern maqāṣid thinkers. However, his understanding of Sharī'ah as a religion that respects human moral agency and social responsibility necessarily incorporates dignity" [5]. Ibn Ashur extended this framework to encompass financial dignity, arguing that exposing an individual's financial weakness to the public constitutes harm to their honour [6]. Classical Islamic sources consider concealed charity (*sadaqah al-sirr*) superior to public giving precisely because it protects the recipient's dignity. A blockchain that permanently records zakat eligibility on Etherscan contradicts this principle directly.

Zero-Knowledge Proofs (ZKPs) address this contradiction by enabling cryptographic verification without exposing the underlying data. Hameed et al. [7] and Wen et al. [8] have surveyed ZKP constructions and zk-SNARK variants suitable for blockchain deployment. Noir [10] was selected over alternatives such as Circom because zakat eligibility rules involve conditional business logic, namely nisab thresholds and hawl holding periods, that require readable and auditable code rather than low-level arithmetic circuit descriptions. Noir compiles to an intermediate representation (ACIR) that Barretenberg's UltraHONK prover [9] turns into a succinct proof, verifiable on-chain but keeping the witness private. Buterin et al. [15] have argued that ZKPs can reconcile privacy with regulatory compliance.

This study employs a Design Science Research (DSR) methodology. DSR is appropriate because the primary contribution is an artifact, namely the ZKT protocol, whose design must be iteratively evaluated against Sharia requirements and performance benchmarks. The protocol implements a Noir circuit encoding nisab and hawl as verifiable predicates, a Pedersen-hash nullifier preventing double-claims without deanonymizing recipients, and a four-tier donor privacy model (Private, Semi-Private, Verified-Private, Public).

This paper investigates three research questions. RQ1: Can zero-knowledge proofs reconcile on-chain zakat verification with the Islamic requirement of donor and recipient privacy? RQ2: What protocol architecture achieves this reconciliation while preventing double-claims through cryptographic nullifiers? RQ3: How does the resulting protocol satisfy nafs protection under maqasid al-shariah?

This paper makes the following contributions: (1) a Noir-based circuit encoding nisab and hawl eligibility verification as verifiable predicates, (2) a Pedersen-hash nullifier mechanism preventing re-claiming without identity exposure, and (3) a four-tier donor privacy model (Private, Semi-Private, Verified-Private, Public).

Section II surveys related work across blockchain zakat systems and ZKP frameworks. Section III presents the DSR methodology. Section IV reports implementation, benchmarks, and security analysis. Section V discusses Sharia compliance and deployment assumptions, and Section VI concludes.

II. Literature Review

Research on blockchain zakat is growing across multiple dimensions: platform implementation, bibliometric analysis, regulatory harmonization, and sustainability assessment. Khan's decentralized platform [2] demonstrated that collection and distribution can function without intermediaries. Nazeri et al. [3] evaluated blockchain features in the Malaysian regulatory context. Khairi et al. [4] extended this work through a full ERC-20 token system with automated disbursement and a permanent audit trail. Nazeri et al. [20] examined the potential and challenges more broadly. All of these systems share a common limitation in that every one records all transactional data on a publicly accessible ledger, making complete donor-recipient transparency available to anyone running an Ethereum node. Beyond these implementation-focused works, Fauzi et al. [19] confirmed through bibliometric analysis that research interest is climbing across Muslim-majority countries, while Ascarya [20] showed that Islamic social finance instruments (zakat, infaq, sadaqah, waqf) supported economic recovery during COVID-19 and Unal and Aysan [21] proposed blockchain as a harmonization tool for differing Sharia-compliance frameworks. Muharam and Osman [26] explored the sustainability dimension. Evaluated through the maqasid al-shariah framework, all of these systems prioritize *hifz al-mal* (protection of wealth) through audit trails and immutable records, but none addresses *hifz al-nafs* (protection of dignity) through donor-recipient privacy. None of these works integrate privacy-preserving transaction mechanisms.

Examined side by side, the three core systems differ in scope but converge on the same privacy posture. Khan's platform [2] operates at the collection-and-distribution layer without specifying a token standard or consent model. Nazeri et al. [3] evaluated blockchain features in the Malaysian regulatory context but did not implement a working system. Khairi et al. [4] produced the most complete implementation with ERC-20 tokens, automated disbursement, and an audit trail, yet none of these works incorporate any mechanism for selective disclosure, donor anonymity, or recipient identity protection. The absence is structural, not incidental, because each system treats full transparency as a design requirement rather than a limitation to be mitigated.

On the ZKP side, the survey literature is extensive across protocol surveys, tooling and frameworks, and application domains. Hameed et al. [7] covered the protocol space and Wen et al. [8] focused on zk-SNARK variants for blockchain deployment. Williams [21] positioned ZKPs within the broader blockchain architecture. On the tooling front, Poseidon, introduced by Grassi et al. [11], reduced constraint counts by up to 8× compared to Pedersen Hash on EVM chains. This work uses Noir [10], a Rust-like DSL that compiles to ACIR, with Barretenberg's UltraHONK prover generating the proof and a Solidity verifier checking it on Sepolia. The pipeline stays client-side, meaning private inputs never leave the donor's machine. Proof hashes anchor on-chain for anyone to audit. Buterin et al. [15] made the broader case that ZKPs can reconcile privacy with regulation, and Mühle et al. [18] applied zk-SNARKs to identity management for KYC.

Three research gaps emerge from this landscape. Existing blockchain zakat platforms, by their own design, lack any privacy-preserving mechanism. Nazeri et al. [3] acknowledged as much when they identified complete traceability as a design objective, rather than a limitation to be overcome. General-purpose ZKP frameworks can anonymize transactions but carry no notion of Islamic eligibility. They hide a transfer without proving the sender meets nisab or hawl. No prior work combines programmable privacy, meaning Noir circuits with verifiable predicates, with Islamic social finance instruments. This paper addresses these gaps.

Fig. 1 positions this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology (bottom-right) justifies the artifact-driven approach of building, evaluating, and refining the protocol on-chain. ZKT bridges these domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, instantiated and validated via Sepolia deployment (v8). These gaps motivate the artifact-driven design science methodology described in the following section.

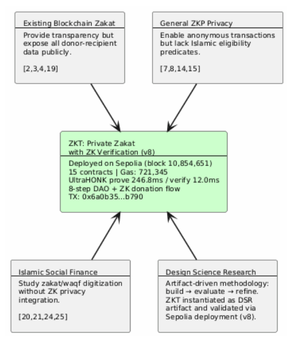

Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

III. Methodology

This study employs a Design Science Research (DSR) methodology, appropriate because the primary contribution is an artifact whose design must be iteratively evaluated against Sharia requirements and performance benchmarks.

The methodology follows five phases as mapped in Fig. 3. Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZKT protocol, the

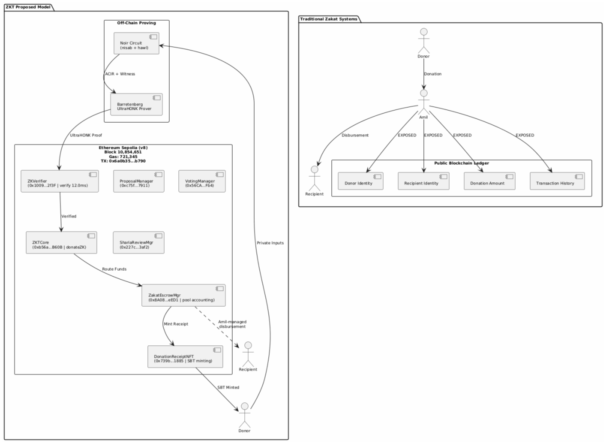

Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

Noir circuit architecture encoding nisab and hawl verification, and the four-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v4.2.1, 15 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 15 PWA frontend with wagmi and XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with 247 ms proof generation and 12.0 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis described in Section IV, and Sharia compliance review conducted with input from Tawf Labs domain experts. RQ1 and RQ2 are addressed through Phases 2–4, and RQ3 through Phase 5.

Validity is limited by single-testnet deployment and single-machine benchmarking. The Sharia review involved manual expert assessment, introducing potential researcher bias. The Sepolia deployment (v8) and benchmark data are available at https://github.com/tawf-labs/zkt-hackathon to support independent replication.

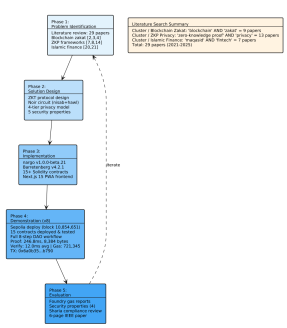

Fig. 3. DSR Methodology with five phases and literature search summary

The DSR methodology produced the results that follow, organized from system architecture to performance evaluation and security verification.

IV. Results

A. System Overview and Threat Model

The ZKT protocol uses a five-actor trust model. The actors are donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a potentially malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

TABLE I
ZKT Protocol Security Properties

Property	Guarantee
Donor anonymity	Identities and amounts never exposed on-chain. Only ZK proofs are publicly verifiable.
Recipient privacy	Donor-side privacy via ZK proofs. Recipient encrypted notes reserved for future work.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data, assuming access to a Sepolia node and independent verification of the Honk verifier contract [5], [6].

For the purposes of this protocol, privacy is operationally defined as the inability of an external blockchain observer to link a donor address to a specific donation amount, a recipient address to a zakat claim, or a donor-recipient pair using only on-chain data. Accountability is defined as the completeness and public verifiability of the `ZKDonationReceived` event stream and the nullifier registry, such that an amil institution can independently reconstruct aggregate zakat flow without accessing individual identity records. Institutional trust (*amanah*) is measured by whether the `ProofVerified` event enables independent audit of every claim processed on-chain.

The protocol is considered successful if four criteria are met. First, no cleartext donor-recipient pairs appear on-chain when the donor selects Private mode. Second, duplicate nullifier submissions are rejected with 100% reliability. Third, proof generation completes entirely client-side, with no private witness data transmitted to any external node. Fourth, amil aggregate compliance reports are independently auditable from emitted events without requiring off-chain data sources.

Fig. 4 shows the three-layer protocol architecture. The off-chain proving pipeline compiles Noir circuits to ACIR and generates UltraHONK proofs via Barretenberg at 247 ms. The Sepolia settlement layer hosts ZKTCore and ZakatEscrowManager. The Next.js PWA frontend uses wagmi/XellarKit for wallet connectivity.

B. System Architecture and Payment Flow

Noir circuits run client-side. Sepolia handles settlement. Donors choose stablecoins or ETH. Everything flows through

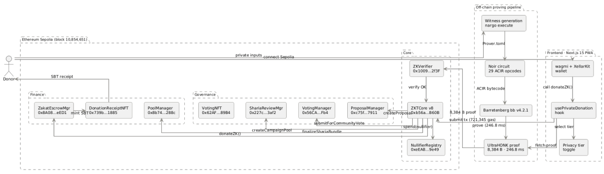

Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

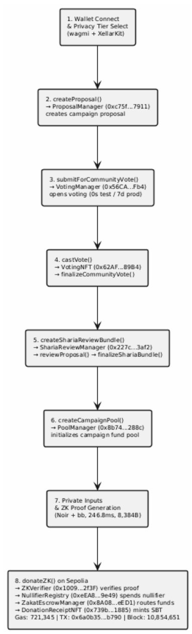

Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

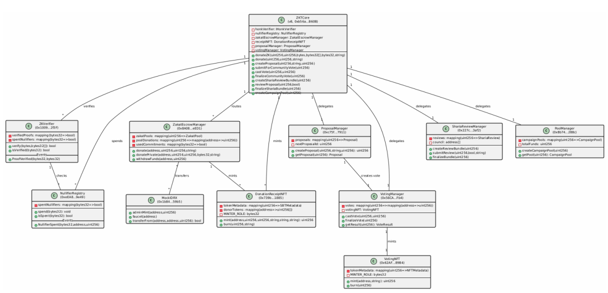

Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

`donateZK()`. Proof generation occurs off-chain, while verification runs on-chain. Privacy is provided by the ZK proofs. Accountability is provided by emitted events and the nullifier registry.

The donor goes through a Progressive Web Application that handles wallet connection, privacy tier selection, donation amount entry, proof generation display, and transaction confirmation. A Web Worker does the actual proving, relying on browser isolation to ensure private inputs never leave the donor's device [11].

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1 on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end donateZK() gas was measured on the Sepolia testnet at block 10,854,651.

TABLE II
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM TARGET, $n = 5$ RUNS

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	246.8 ms
Proof verification (avg)	12.0 ms
Proof size	8,384 bytes

D. Sepolia Deployment and Gas Consumption

15 contracts deployed via single forge script to Sepolia (chain ID 11155111). The ZKTCore (v8, 0xb56a...B60B) routes `donateZK()` through `zakatEscrowManager.donate()` for pool accounting. An end-to-end donation (tx 0x6a0b...b790, block 10,854,651) completed proof verification, nullifier registration, transfer, NFT minting, and event emission in 721,345 gas.

Table III reports gas costs measured via Foundry gas reports.

TABLE III
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT, ETHEREUM SEPOLIA

Contract	Function	Gas (avg)
ZKTCore	`donate()`	490,187
ZKTCore	`createCampaignPool()`	365,552
ZKTCore	`castVote()`	132,909
ZKTCore	`donateZK()` (e2e)	721,345
ZKTCore	Deploy	5,314,471
ZakatEscrowMgr	Deploy	3,069,194
ShariaReviewMgr	Deploy	2,114,319

The gas measurements establish the protocol's on-chain cost. Beyond performance, the protocol must satisfy specific security guarantees. Each is analyzed below.

E. Security Analysis

Donor anonymity. The only on-chain observable from a private donation is the event emitted by `ZKTCore.sol` (lines 577-582):

```
event ZKDonationReceived(
    uint256 indexed poolId,
    bytes32 indexed nullifier,
    uint256 amount,
    address indexed donor
);
```

Private mode zeroes both `donor` and `amount`. Nothing traceable. Semi-Private attaches the donor's address but hides the amount, a design tradeoff for institutional accountability. Neither mode leaks the full (donor, amount) pair. The verifier contract checks only a hash and a nullifier flag. These two fields constitute the complete on-chain footprint of a private donation.

Recipient privacy. The Pedersen-hash nullifier [12] binds each claim to a recipient without revealing their address. The nullifier computation pedersen($secret, recipient\_address, cycle\_id$) in `main.nr` means the on-chain nullifier is deterministic but preimage-resistant. An observer sees `0x3f2a...` and cannot recover the recipient address or secret. Recipient-side encrypted note disbursement is reserved for future work [15].

Circuit soundness. Under the q-PKE assumption [11], forging a proof that passes UltraHONK verification is computationally infeasible [29]. A non-eligible recipient, whose wealth exceeds the nisab threshold as checked by `verify_nisab()` in `main.nr`, cannot produce a witness that satisfies the circuit constraints.

Double-donation prevention. The on-chain `NullifierRegistry` maintains a `mapping(bytes32 => bool)` of spent nullifiers. The `spend()` function enforces single-use through `require(!spentNullifiers[nullifier])`, rejecting any nullifier already recorded. The `ZKTCore.donateZK()` function atomically chains `honkVerifier.verify()` → `nullifierRegistry.spend()` → `zakatEscrowManager.donate()`. A duplicate nullifier causes the entire transaction to revert. Each eligible recipient can claim zakat at most once per cycle. The nullifier hash is preimage-resistant, so an observer cannot derive the recipient address from it, but the address itself is a public circuit input visible in the transaction calldata. Full recipient privacy, where not even the amil institution can identify claimants, depends on encrypted note disbursement reserved for future work.

Privacy Exposure Analysis. The current prototype exposes several privacy vectors beyond those resolved by the nullifier mechanism. The recipient address is a public circuit input transmitted in the calldata of every `donateZK()` transaction, making it visible to any blockchain observer and to the amil institution. In Semi-Private mode, the donor address is emitted in the `ZKDonationReceived` event. Transaction graph analysis on the public Sepolia ledger could potentially correlate donation timing with recipient claim timing. The Web Worker proving model isolates private witness data from the browser main thread but does not protect against IP-level or browser-based side-channel attacks. Donations to the same pool reveal aggregate economic patterns even when individual amounts are hidden. Each of these exposures is acknowledged as a limitation of the current prototype and is addressed in the future work roadmap through encrypted note disbursement [15], private decentralized identifiers, and potential integration with on-chain mixer patterns [16].

The security analysis confirms that the protocol meets its design goals. The following section discusses these results in the context of Sharia compliance and deployment feasibility.

V. Discussion

A. Sharia Compliance and Deployment

Maqasid al-shariah is anchored by five *daruriyyat*, namely religion, life, intellect, progeny, and wealth [5]. Al-Shatibi formulated these categories centuries before blockchain existed. Ibn Ashur later extended *hifz al-nafs* to encompass dignity, arguing that exposing an individual's financial weakness to the public constitutes harm to their honour [6]. A single address recorded as a zakat recipient on Etherscan broadcasts below-nisab status permanently, an outcome that aligns directly with Ibn Ashur's reasoning even though the technology itself was unimaginable in his time.

Azizov et al. [25] propose a maqasid framework for fintech regulation. Latang et al. [24] examine the Islamic legal status of cryptocurrency assets. Pramuka et al. [27] explore blockchain's role in Islamic charitable crowdfunding. This work addresses the donor side, using ZK proofs and nullifier-based anonymity. Recipient-side privacy through encrypted private notes is reserved for future work.

Two maqasid dimensions are directly engaged. *Hifz al-nafs* manifests through Private and Semi-Private modes keeping donations and linkages off-chain. *Hifz al-mal* operates through the nullifier registry stopping double-claims and the proof's soundness blocking fake eligibility. *Amanah*, institutional trust, flows from the `ProofVerified` event. Amil institutions audit aggregate flows without ever touching individual records.

Production deployment requires three prerequisites. Indonesia's OJK Fintech Sandbox must grant regulatory approval, amil institution staff must be trained on wallet management and proof verification workflows, and a rupiah-backed digital currency (such as Bank Indonesia's planned CBDC under Project Garuda) must replace the mock IDRX stablecoin.

B. Limitations and Future Work

The circuit covers nisab and hawl. Eight asnaf categories are architecturally possible but not yet implemented. The amil institution disburses through ZakatEscrowManager today with no on-chain proof the money reached the mustahik. DIDs for recipients would close that gap and make disbursements verifiable. A dual governance model, Sharia Council ZK DAO plus Community Donor DAO, is under design. Other directions include recursive proof systems for multi-period eligibility, compliance-friendly ZK protocols for AML/CFT, and batched settlement optimization [7], [17], [23]. These limitations frame the concluding summary of the protocol's principal findings and its path toward production deployment.

VI. Conclusion

A privacy-preserving zakat protocol was deployed on Ethereum Sepolia using Noir circuits and UltraHONK proofs. The circuit (29 ACIR opcodes) verifies nisab and hawl eligibility client-side, a Pedersen-hash nullifier prevents double-claims without deanonymizing recipients, and a four-tier privacy model lets donors choose their disclosure level. Across five benchmark runs, proof generation averaged 246.8 ms (Barretenberg v4.2.1) and an end-to-end Sepolia transaction consumed 721,345 gas.

The core finding is that cryptographic privacy and Sharia compliance are not in tension. Ibn Ashur's framing of *nafs* protection as encompassing financial dignity [6] aligns with ZK proofs, which verify eligibility while keeping identity, income, and assets private. The `ZKDonationReceived` event gives amil institutions an auditable record of aggregate zakat flow, while the nullifier registry prevents abuse, decoupling what the blockchain records from what it reveals about individuals.

Future work extends the circuit to all eight asnaf categories, adds recipient-side encrypted note disbursement, and pursues OJK Fintech Sandbox integration for production deployment.

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